

1.1 What is economics?

Learning objectives

1.1 What is economics?	Depth	Diagrams & calculations
Economics as a social science • The social nature of economics • The basis of the study of economics: microeconomics and macroeconomics • Introduction to the nine central concepts: scarcity, choice, efficiency, equity, economic well-being, sustainability, change, interdependence, intervention	AO2	
The problem of choice • Factors of production—land, labour, capital and entrepreneurship • Scarcity • Unlimited human needs and wants to be met by limited resources • Scarcity and sustainability	AO2	

Learning objectives

1.1 What is economics?	Depth	Diagrams & calculations
• Opportunity cost • The cost of choice • Free goods • The basic economic questions • What/how much to produce, how to produce and for whom to produce? • Means of answering the economic questions • Market versus government intervention • Economic systems: free market economy, planned economy and mixed economy	AO2	

Learning objectives

1.1 What is economics?	Depth	Diagrams & calculations
The production possibilities curve model (PPC) • Assumptions of the model • Increasing versus constant opportunity cost • Features of the model: opportunity cost, scarcity, choice, unemployment of resources, efficiency, actual growth and growth in production possibilities	AO2 AO4	Diagram: PPC illustrating choice and opportunity cost, unemployment of resources, actual growth and growth in production possibilities Diagram: PPC showing increasing versus constant opportunity cost

Learning objectives

1.1 What is economics?	Depth	Diagrams & calculations
Modelling the economy • The circular flow of income model • Interdependence between economic decision-makers interacting and making choices in an economy: households, firms, the government, the banks and financial sector, and the foreign sector (foreign firms and households) • Leakages and injections	AO2 AO4	Diagram: circular flow of income model, with leakages and injections

What is economics?

Economics is a **social science** which studies the behaviour of individuals and societies when allocating scarce resources to meet unlimited needs and wants.

Economics uses **scientific methods** to model the behaviour and interactions between economic agents such as consumers, producers, and the government.

Brainstorm – what did we learn last lesson?
In pairs, write down as much as you can remember.

Economics is a social science

Microeconomics

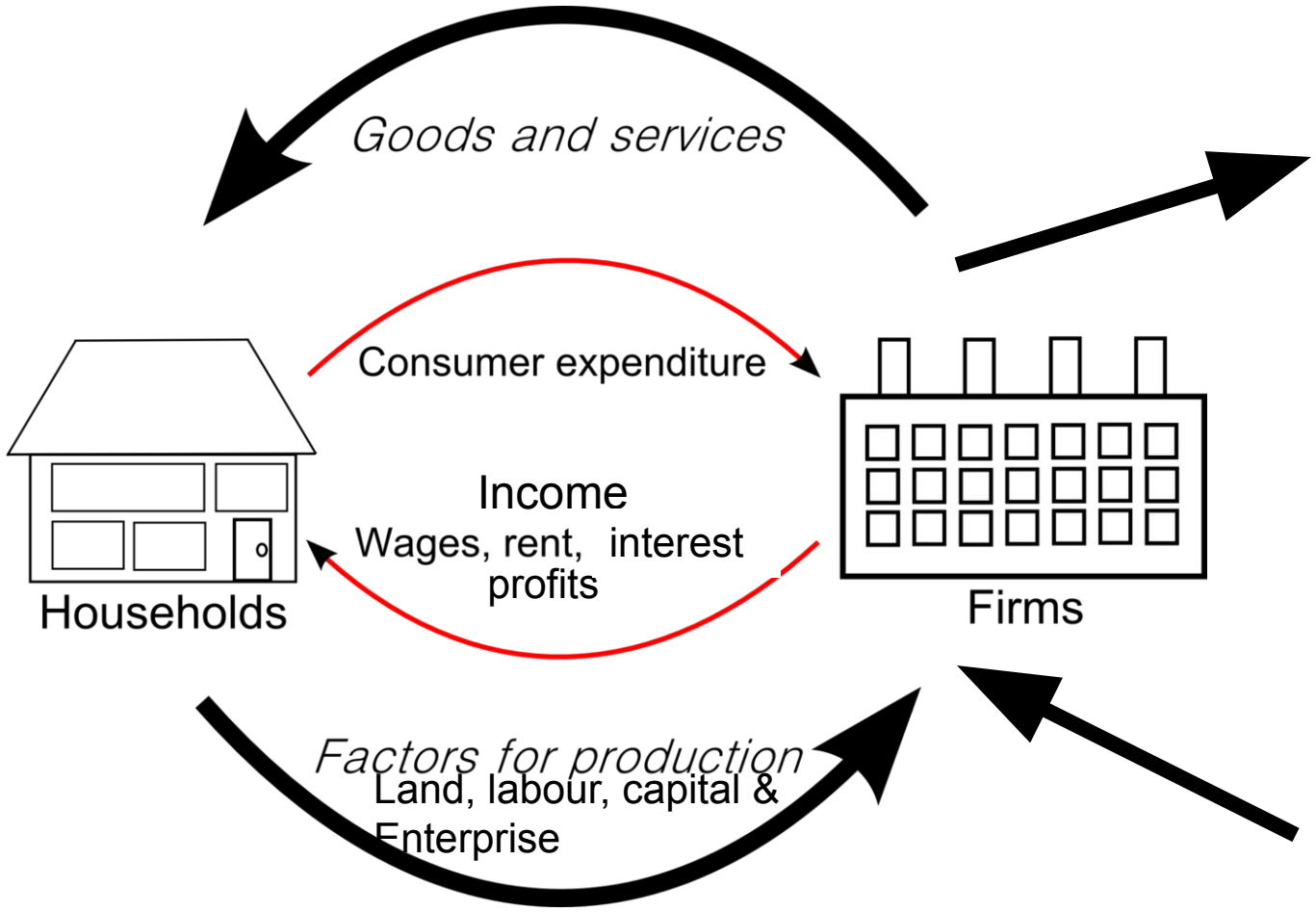
Marcroeconomics

Factors of production:
Land, Labour, Capital and Enterprise

Nine key concepts: **WISE ChoICES**

Well-being
Interdependence
Scarcity
Efficiency
Choices
Intervention
Change
Equity
Sustainability

The Circular Flow of Income



Withdrawals/ leakages

- Saving
- Imports
- Taxes

Injections

- Investment
- Exports
- Government Expenditure



Factors of production

Factors of production refer to the resources that are required to produce goods and services.

The four factors of production can be remembered using the acronym **CELL**.



Factors of production

Capital refers to man-made resources e.g., vehicles, machinery, and infrastructure.



Factors of production

Enterprise refers to the skill of organizing the three other factors of production. Famous entrepreneurs include Elon Musk, Jeff Bezos, Oprah Winfrey, and Vera Wang.



Factors of production

Land refers to natural resources e.g., agricultural land, wood, gold, water, and oil.



Factors of production

Labour refers to human resources e.g., factory workers, teachers, artists, and engineers.

Factors of production – factor income

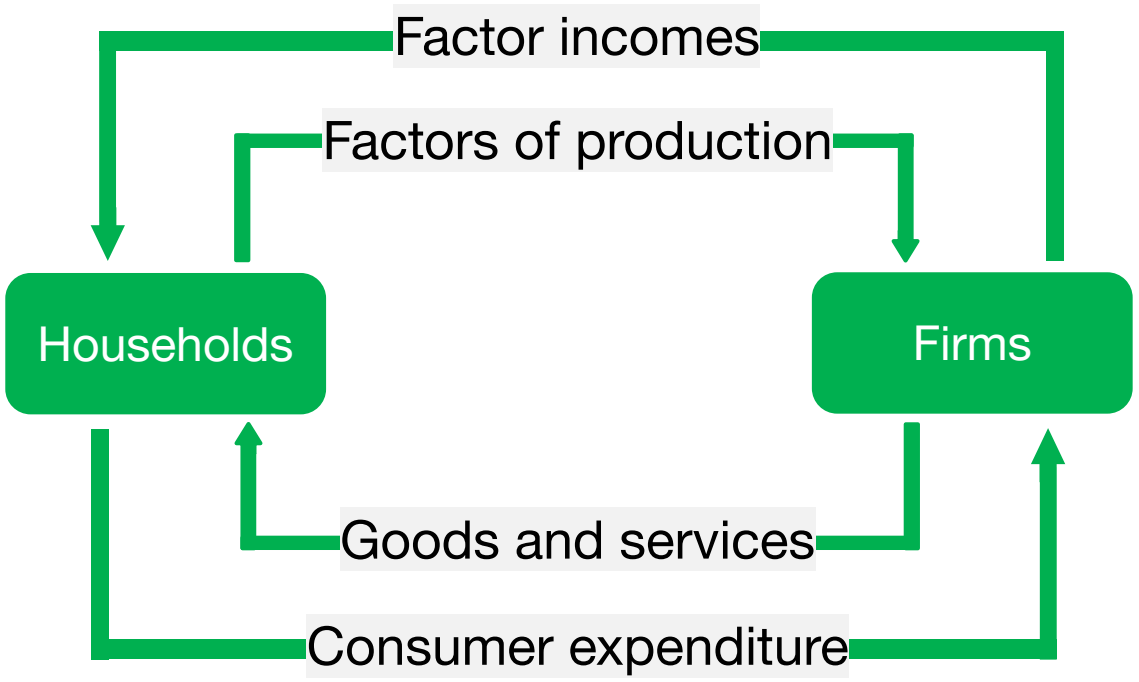
Households own the factors of production which are sold to firms in exchange for income.

Factor of production	Factor income
Capital	Interest
Entrepreneurship	Profit
Land	Rent
Labour	Wages

Circular flow of income

The **circular flow of income** model shows the interaction between households and firms.

- Households provide resources to businesses.
- Firms provide income to households.
- Firms use the resources to produce goods and services for households.
- Households provide expenditure to businesses.
- The cycle repeats.



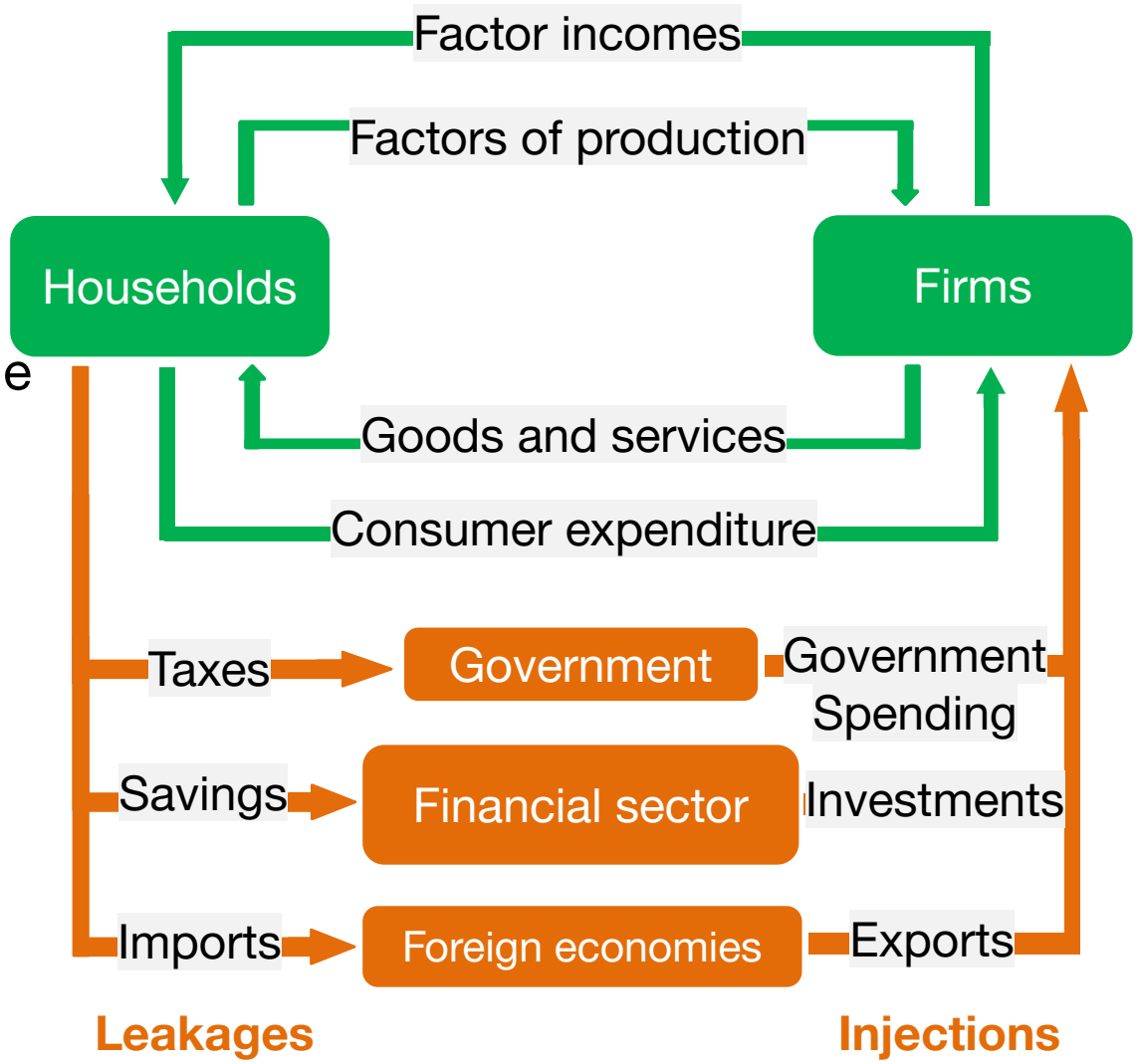
The circular flow of income model

An **injection** refers to the additional money added into the circular flow of income.

A **leakage** refers to a withdrawal of money from the circular flow of income.

If **injections** > **leakages**, circular flow increases.

If **leakages** > **injections**, circular flow decreases.





Real world example – Research task

Step 1: Visit World's most expensive cities

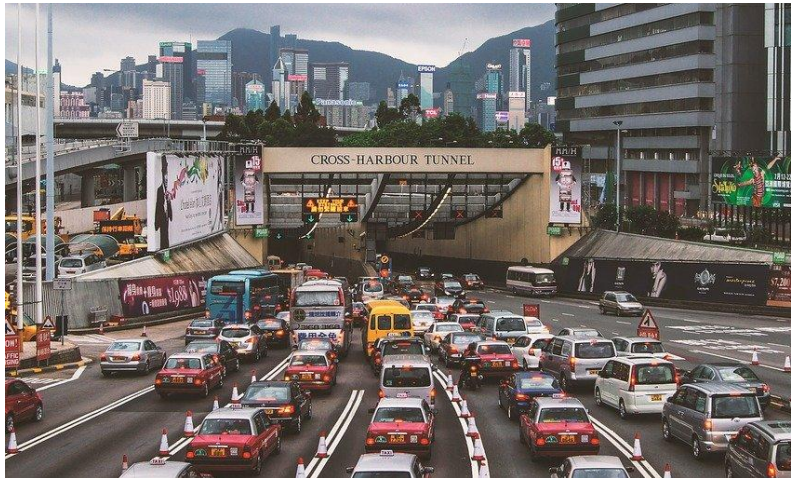
Step 2: Complete tasks on the next slide

Real world example – research task

Source: World's most expensive cities

1. Sort the table by Buying Price (US$ per sqm)
2. Using examples, consider why property prices in some cities are much higher than others.
3. Which of the 9 key concepts does this exercise relate to the most? Justify your answer.







Scarcity & the economic problem

There are **limited** resources in the economy, but individuals and societies have **unlimited** wants.

Therefore, **choices** must be made about how to allocate scarce resources.

Economic systems aim to answer the three basic economic questions:

1. What to produce?
2. How to produce?
3. For whom to produce for?



Consider the similarities and differences between how different economies address these three basic economic questions.

Real world example

Source: Peak Population: What it Means for Global Resources

1. What are finite resources?
2. List three examples of finite resources.
3. List some of the factors which may affect the carrying capacity of a population.
4. List some of the issues which may arise due to overpopulation and the scarcity of resources.



Economic systems

Free market economy	Mixed economy	Planned economy



Allocation of resources is determined by market forces i.e., consumers and producers.	Allocation of resources is determined by consumers, producers, and the government.	Allocation of resources is determined by the government.

Limited resources and **unlimited wants** leads to the problem of **scarcity**, which means that **choices** have to be made.

Opportunity cost – the cost of choice

Every choice made is associated with a cost, which is known as an opportunity cost.

Opportunity cost refers to the value of the next best alternative forgone when a choice is made.

Examples:

- A worker decided to go on vacation for a week instead of going to work. The opportunity cost is the lost wages.
- The US government spent over $1 trillion on social security in 2020. The opportunity cost is $1 trillion that could have been used to improve education, healthcare, or other areas of the economy.







Opportunity cost – free goods

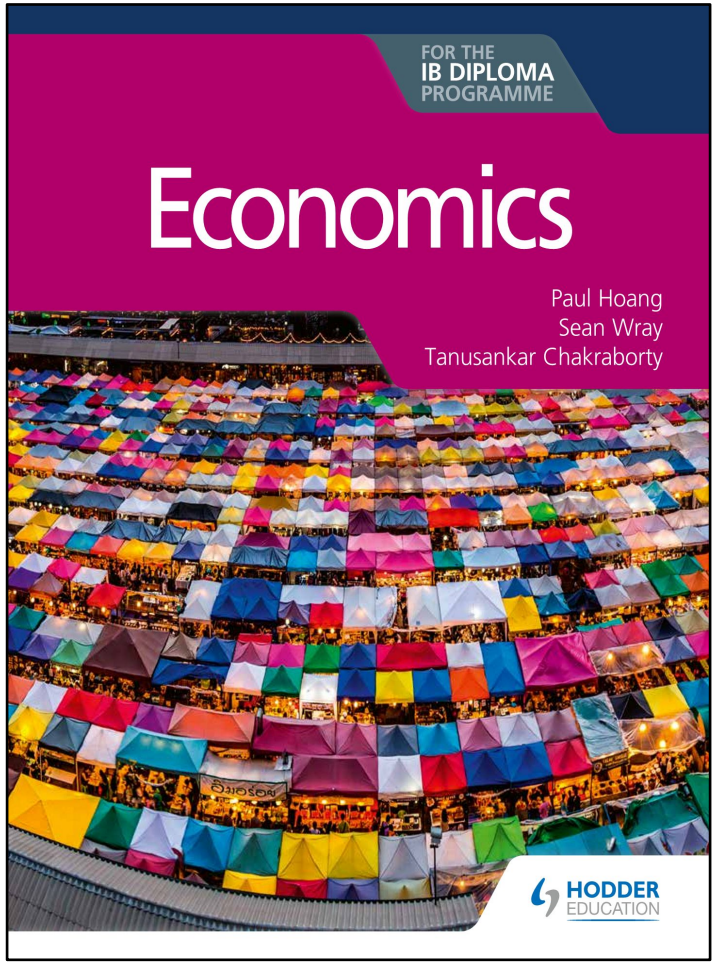
Free goods are naturally abundant with unlimited supply and therefore do not incur any opportunity costs. Examples include air, sunlight, sea water, and desert sand.

Over to you…

Hoang, Wray, & Chakraborty (2020)

Economics for the IB Diploma Programme

- Pages 14 –15
- Paper 2 and 3 Exam Practice Questions 1.1 and 1.2



Over to you…

PAPER 2 AND 3 EXAM PRACTICE QUESTION 1.1

Consider the table below, which shows the maximum combination of consumer and producer goods in an economy:

Consumer goods		Producer goods
100 units	plus	50 units
80 units	plus	60 units

Calculate the opportunity cost of one unit of producer goods in terms of units of consumer goods. *[2 marks]*

Over to you…

PAPER 2 AND 3 EXAM PRACTICE QUESTION 1.2

A farmer has land on which carrots and corn are grown. The production possibilities are shown in the table below:

Carrots ('000 kg)	Corn ('000 kg)
14	4
10	10
6	16
2	22

Calculate the farmer's opportunity cost of producing 1 kilogram of carrots in terms of kilograms of corn. *[2 marks]*

Over to you…

Chapter 1 – What is economics?

[Paper 2 and 3 Exam practice question 1.1]

By producing an extra 10 units of producer goods, the opportunity cost is 20 units of consumer goods, i.e. a ratio of 10:20 or 1:2. Therefore, the opportunity cost of producing 1 extra unit of producer good is 2 units of consumer goods.

Award *1 mark* for the correct answer, and 1 mark for appropriate working out.

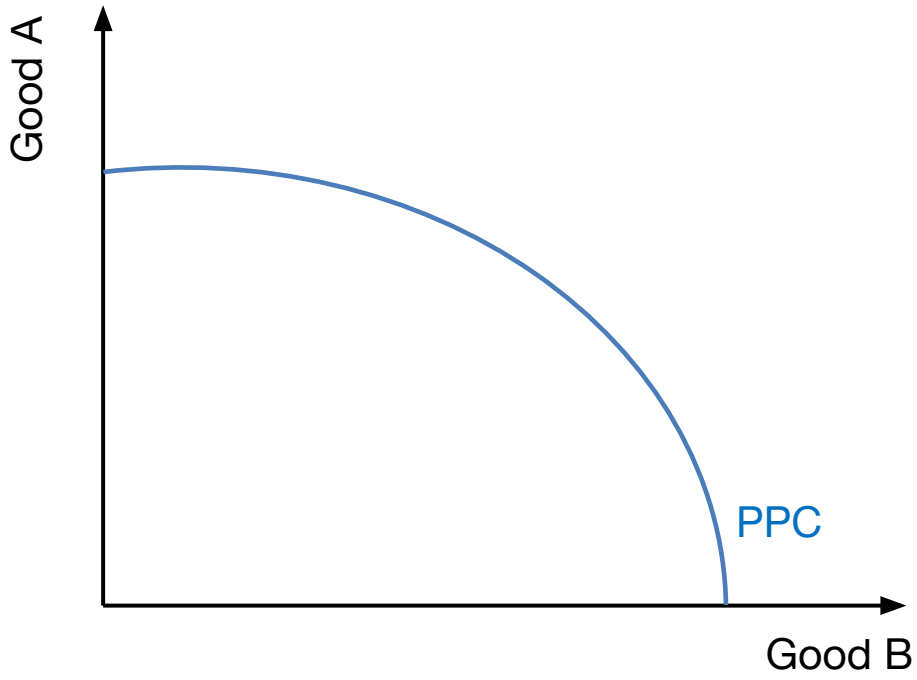
[Paper 2 and Paper 3 Exam practice question 1.2]

For every additional 4,000 kilograms of carrots, the farmer sacrifices 6,000 kilograms of corn, i.e. a ratio of 4:6 or 1:1.5. This means that the opportunity cost of producing 1 kilogram of carrots is 1.5 kilograms of corn..

Award *1 mark* for the correct answer, and *1 mark* for appropriate working out.

The production possibilities curve model (PPC)

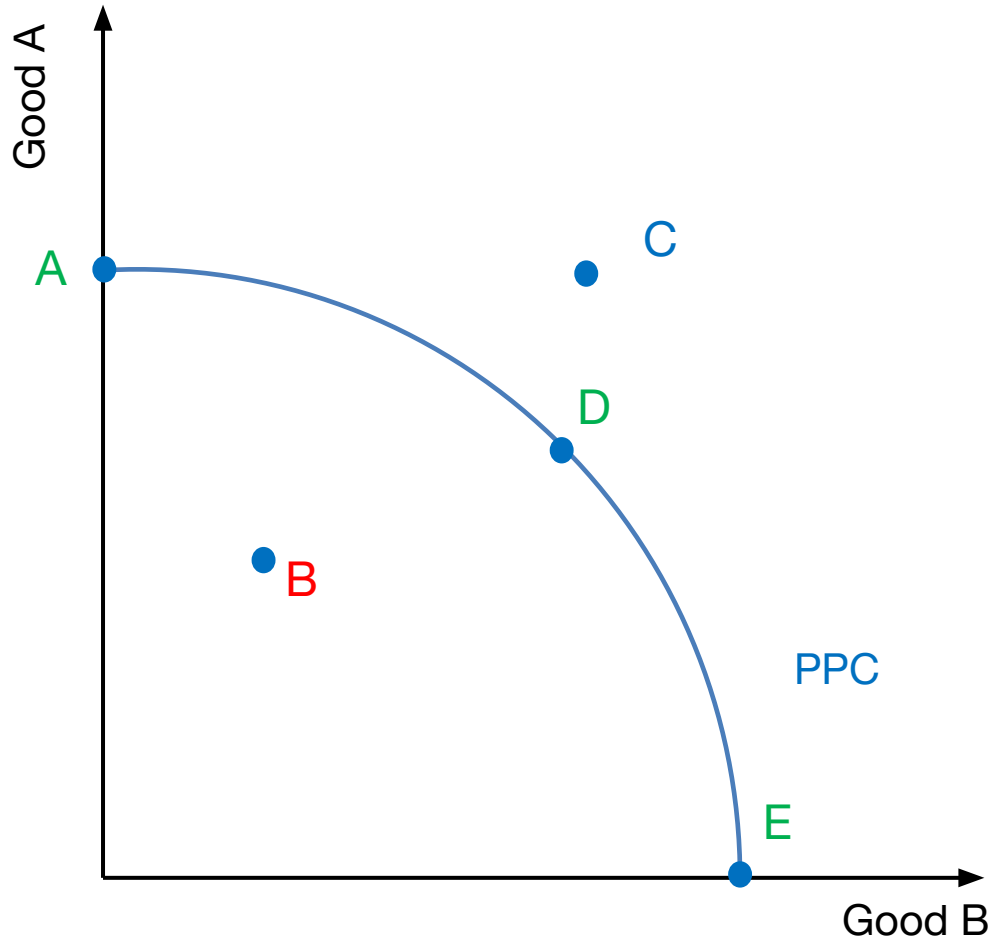
The **production possibilities curve** is a model which illustrates the different combinations of two goods or services that an economy or firm can produce.



Assumptions of the model:

- There is a fixed amount of resources
- Only two goods are being produced
- Technology and production techniques are fixed
- All resources are used efficiently

The production possibilities curve model (PPC)



A point **on** the PPC (**A, D, E**) means all resources are employed and are used efficiently.

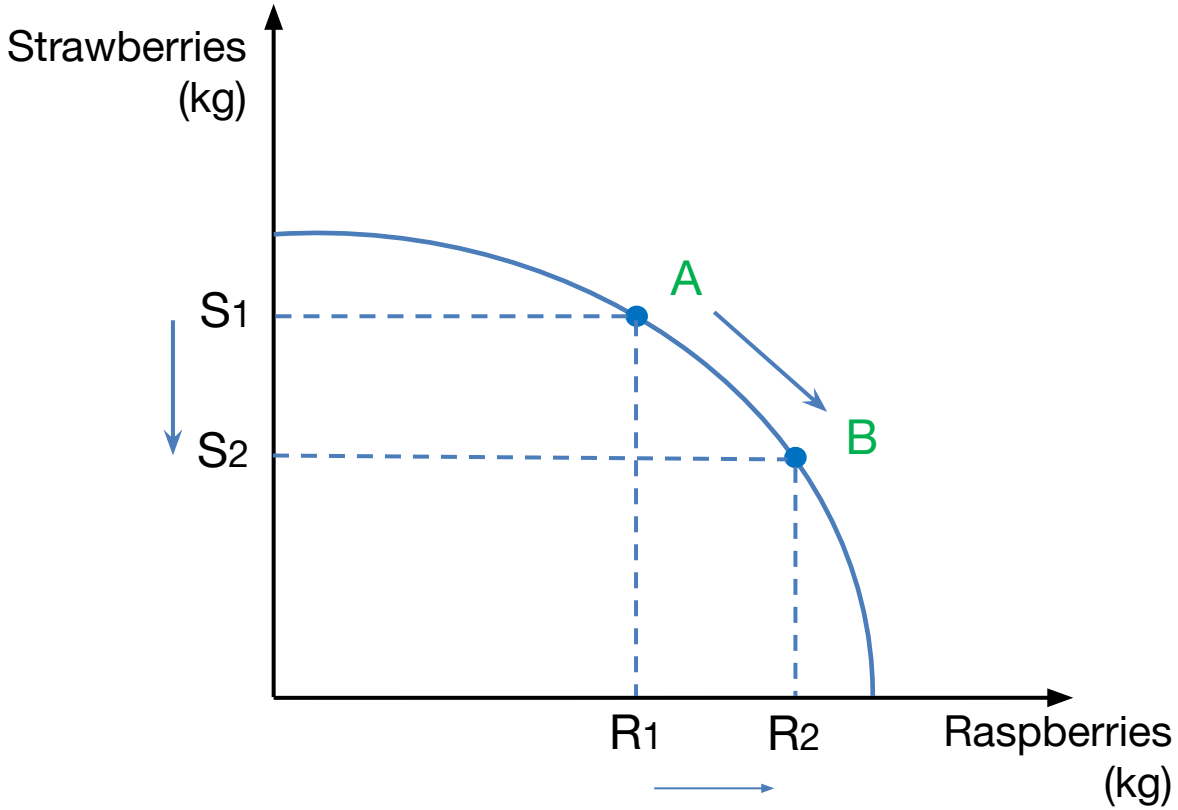
A point **within** the PPC (**B**) means some resources are unemployed or used inefficiently.

A point outside the PPC (**C**) is not possible with the current level of resources.

The production possibilities curve model (PPC)

PPC and opportunity costs

Suppose a farmer decides to change from point A to point B. What is the opportunity cost?

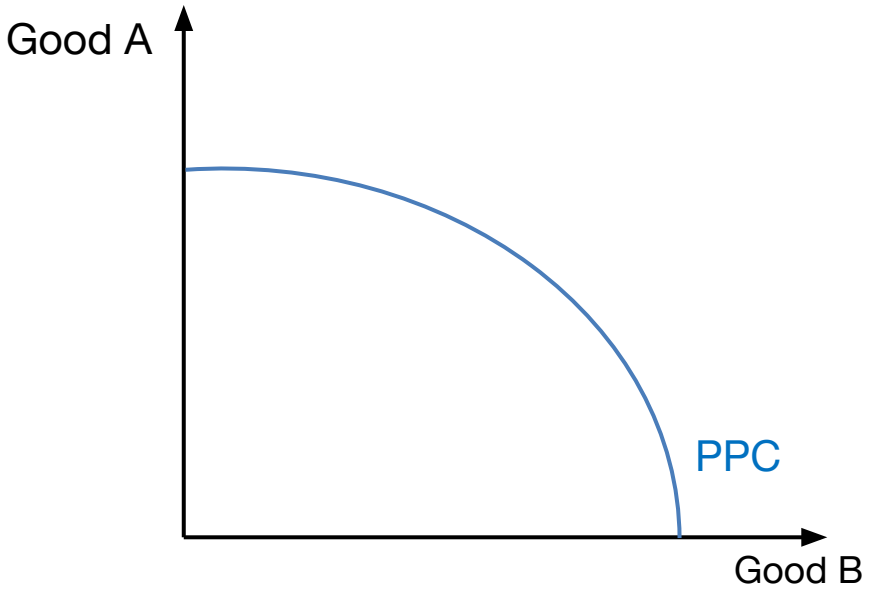


The opportunity cost of producing R2-R1 additional kilograms of raspberries is S1-S2 kilograms of strawberries. This is the opportunity cost of the choice made due to scarce resources such as land and labour.

The production possibilities curve model (PPC)

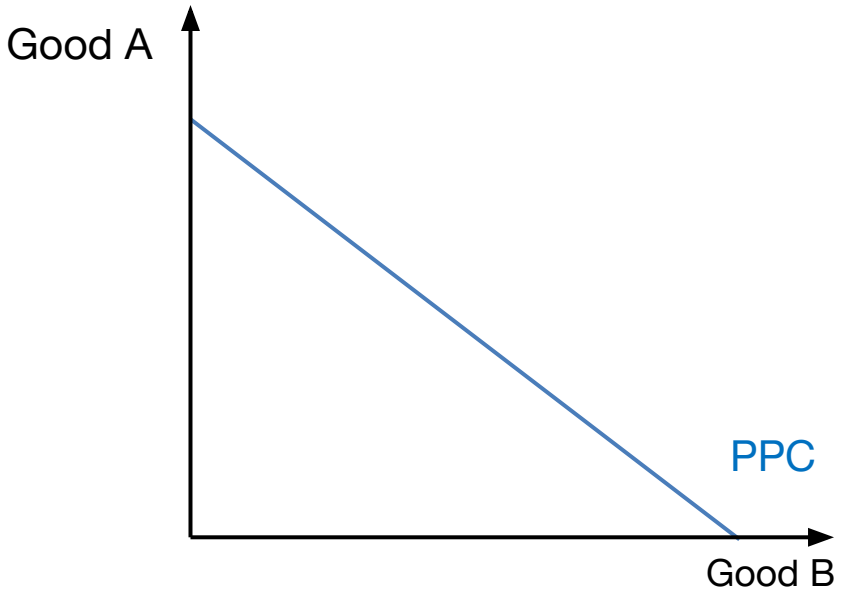
Increasing opportunity cost

A concave PPC occurs due to specialization of resources, as resources are not equally suitable to produce different products.



Constant opportunity cost

A linear PPC occurs due to the marginal rate of transformation being constant throughout the curve.



The production possibilities curve model (PPC)

Constant opportunity cost



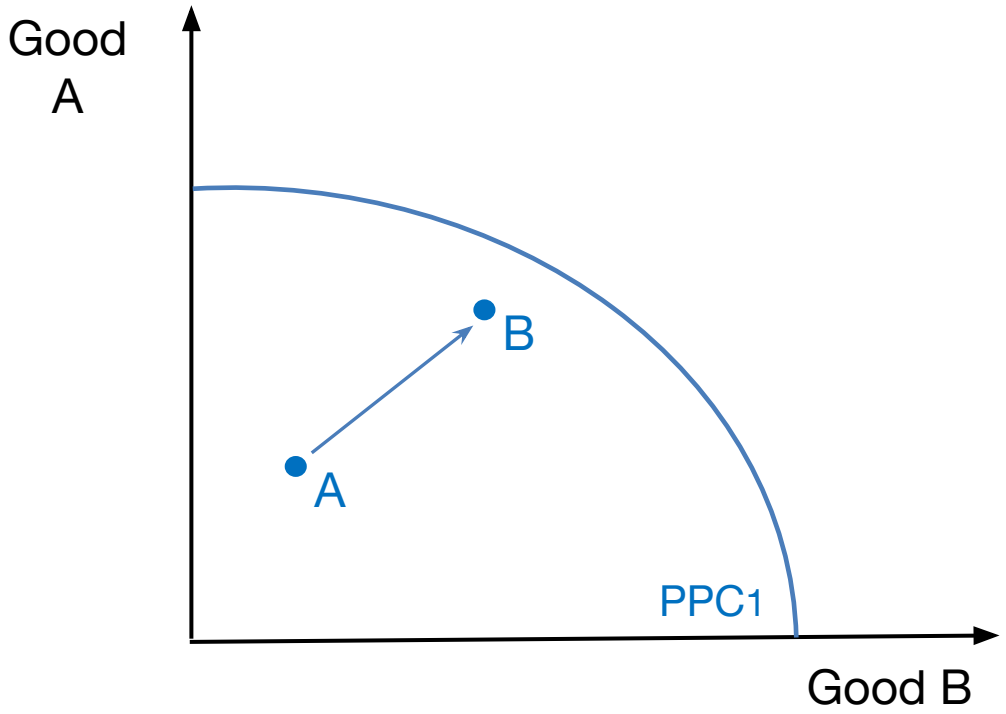
Cheryl finishes school at 4pm and goes to bed at 10pm. She is free to study and play during the hours in between. Assume Cheryl is only interested in studying and/or playing.

Draw a PPC diagram to illustrate the different combinations of hours Cheryl can spend studying and/or playing.

The production possibilities curve model (PPC)

Actual growth (e.g., Point A to Point B)

An economy moves closer to the PPC when unemployment decreases, and efficiency increases.



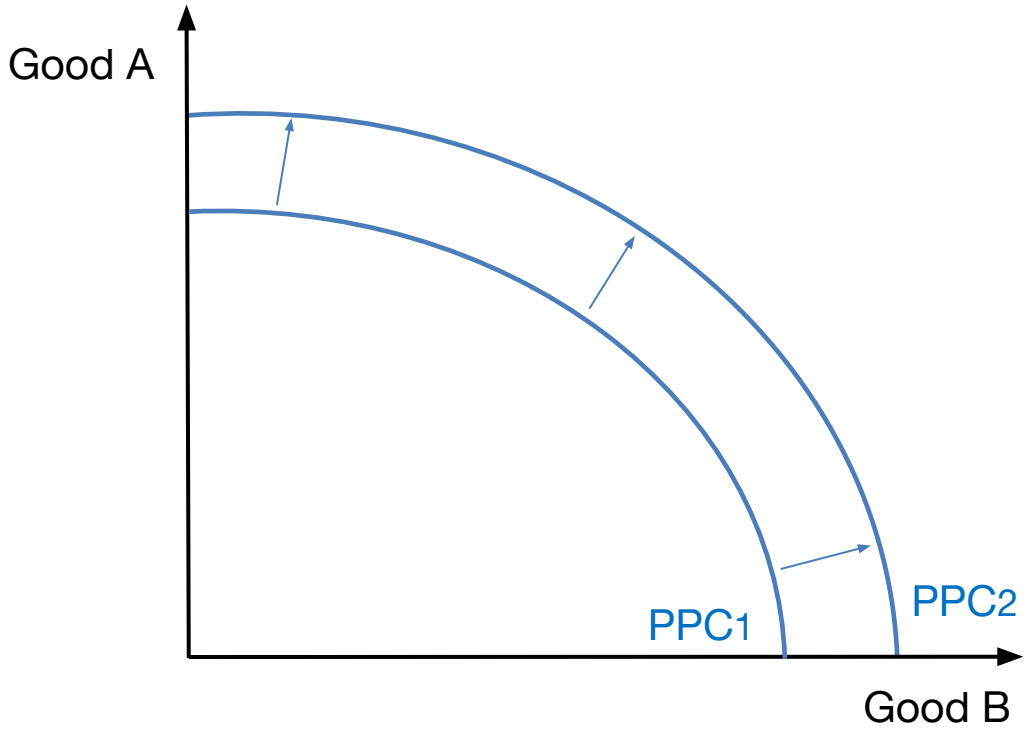
The production possibilities curve model (PPC)

Growth in production possibilities (e.g., PPC_1 to PPC_2)

An economy's PPC can shift outwards if there is:

- an increase in the **quantity** of resources
- an improvement in the **quality** of resources
- an improvement in **technology**

An outward shift of the PPC indicates that the productive capacity of the economy has increased.



Top tip!

You need to be aware of the significance of the location of production possibilities, and be able to explain points under, on and beyond a PPC. The key to the production points in a PPC diagram is outlined below:

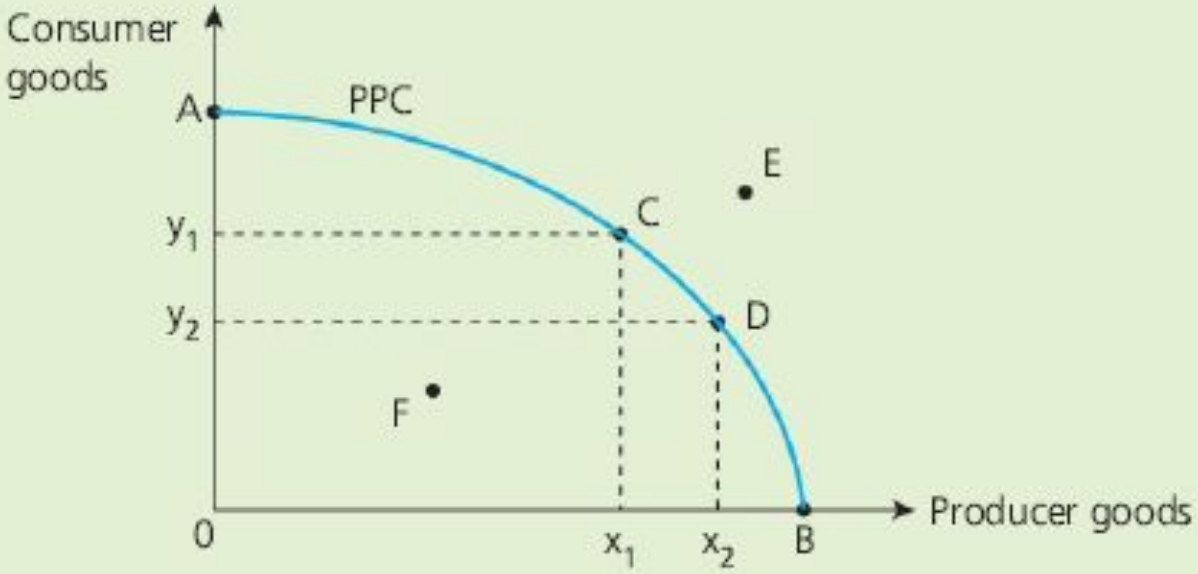


Figure 1.27 The Production Possibility Curve

- Point A – All resources are dedicated to the production of consumer products.
- Point B – All resources are dedicated to the production of producer products.
- Point C – y_1 units of consumer products are produced along with x_1 units of producer products.
- Point D – y_2 units of consumer products are produced along with x_2 units of producer products.
- Point E – This point is beyond the production possibility curve and lies outside the productive capacity of the economy, so it is currently unattainable.
- Point F – This point is within the productive capacity of the economy, so the production of both consumer products and producer products can increase (towards point C or D, for example) without any opportunity cost as some factors of production are currently not being used.

Top tip!

The PPC model is a very important one in the study of economics and can be applied in many areas of the discipline. Notice also that the features of the PPC model include several key **WISE ChoICES** concepts: scarcity, efficiency and choice. Although other concepts such as interdependence, change and economic well-being are not explicitly featured in this model, these concepts can also be seen in a PPC diagram.

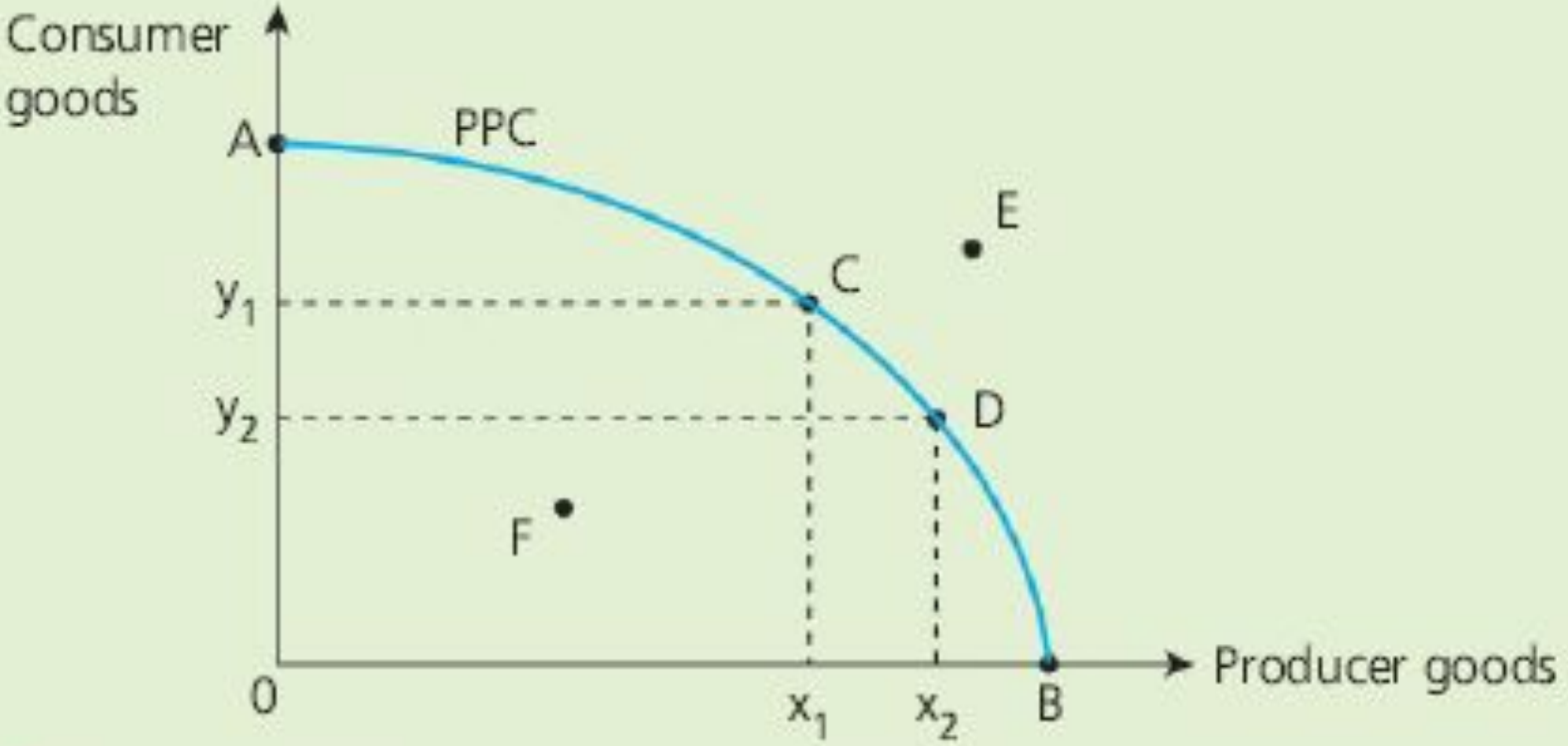


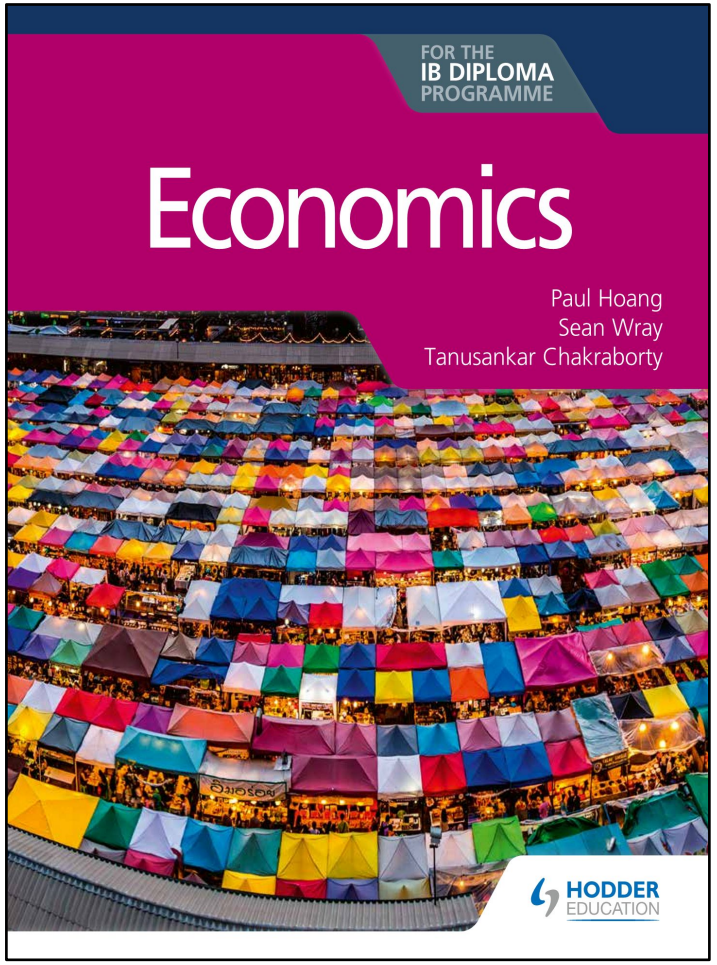
Figure 1.27 The Production Possibility Curve

Over to you…

Hoang, Wray, & Chakraborty (2020)

Economics for the IB Diploma Programme

- Page 24
- Paper 2 and 3 Exam Practice Question 1.3



Over to you…

PAPER 2 AND 3 EXAM PRACTICE QUESTION 1.3

Low-lying areas of Bangladesh are prone to flooding each year. Crops are lost and thousands of people lose their homes. During times of severe flooding, roads and railways are damaged and farmers find it impossible to transport their dwindled crops to market.

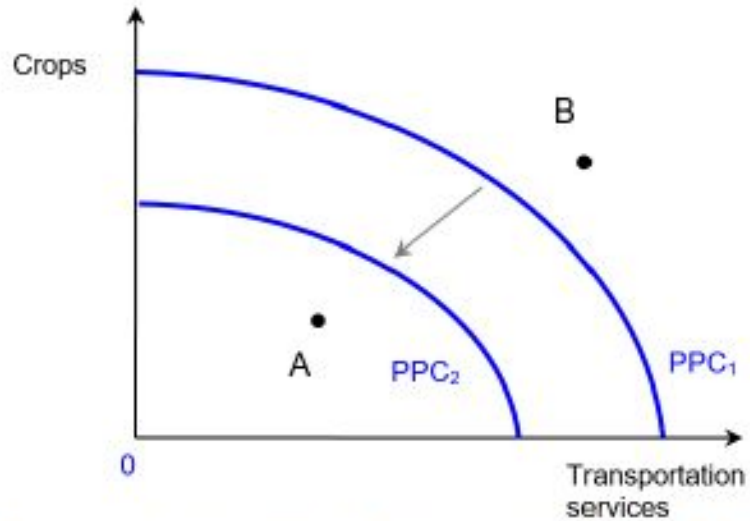
a Draw a relevant PPC diagram and explain the impact of flooding on the productive capacity of Bangladesh. *[4 marks]*

b On the diagram, draw and label a point where some of the factors of production will be idle. *[1 mark]*

c On the diagram, draw and label a point that is unattainable. *[1 mark]*

[Paper 2 and Paper 3 Exam practice question 1.3]

a) PPC diagram for Bangladesh:



The PPC curve of Bangladesh will shift inwards from PPC_1 to PPC_2 because the flooding causes disruptions to transportation networks and damage to crops and homes.

Award up to *2 marks* for drawing and labelling a correct diagram.

Award up to *2 marks* for a full explanation of the diagram.

a) At point A, some of the factors of production are idle (unemployed or underutilized) because this point is positioned inside of the PPC.
b) Point B is currently unattainable because it is positioned outside the PPC.



Test your knowledge on this unit: Kahoot!

1. Using a PPC, explain the concept of opportunity cost.
2. Using a PPC, explain the impact of Vietnam discovering significant oil supplies off its coast.
3. Using a PPC, explain the impact of an increase in net inward migration to a country.
4. Using a PPC, explain the concept of scarcity.
5. Using a PPC, explain the concept of choice.

1. Explain the effects of an increase in exports and an increase in imports on the circular flow of income.
2. Explain the relationship between households and firms in the circular flow of income.

10 mark answers:

- Define key terms
- Draw, fully and explain diagram
- Explain
- Include examples, where relevant

Assessment Criteria

Part (a) 10 marks

Marks	Level descriptor
0	• The work does not reach a standard described by the descriptors below.
1–2	• The response indicates little understanding of the specific demands of the question. • Economic theory is stated but it is not relevant. • Economic terms are stated but they are not relevant.
3–4	• The response indicates some understanding of the specific demands of the question. • Relevant economic theory is described. • Some relevant economic terms are included.
5–6	• The response indicates understanding of the specific demands of the question, but these demands are only partially addressed. • Relevant economic theory is partly explained. • Some relevant economic terms are used appropriately. • Where appropriate, relevant diagram(s) are included.
7–8	• The specific demands of the question are understood and addressed. • Relevant economic theory is explained. • Relevant economic terms are used mostly appropriately. • Where appropriate, relevant diagram(s) are included and explained.
9–10	• The specific demands of the question are understood and addressed. • Relevant economic theory is fully explained. • Relevant economic terms are used appropriately throughout the response. • Where appropriate, relevant diagram(s) are included and fully explained.

Assessment Criteria

Part (a) 10 marks

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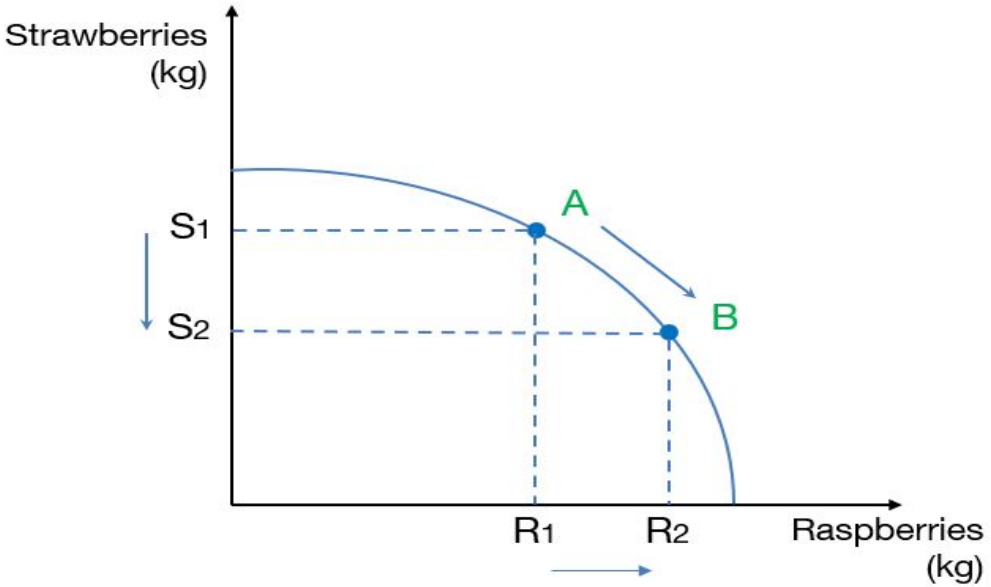
1. Using a PPC, explain the concept of opportunity cost.

Opportunity cost if defined as the next best opportunity forgone. This means that in making a choice individuals, consumers, firms or businesses have given up an alternative option. For example, a student who chooses to spend their lunch money on pizza cannot also spend that same money on pasta. In choosing the option to buy the pizza they have given up the option to buy the pasta.

Opportunity cost arises due to scarcity. While resources are limited human needs and wants are unlimited. This leads to the requirement to make choices over how limited resources will be used and distributed.

A Production Possibilities Curve diagram illustrates this concept. It consists of two axis which represents the potential output of two products or groups of products. The curve is drawn with the assumption that resources are limited, that they are used efficiently, that all resources are fully employed, and a given level of technology.

To increase the output of one good more resources are needed. If these resources are used to increase output of good A then there are less resources to produce good B. Therefore, to product more of good A some of good B has to be given up. This is known as the opportunity cost.



For example, a farmer choosing how much land and labour to use for strawberry and raspberry production. The opportunity cost of producing R2-R1 additional kilograms of raspberries is S1-S2 kilograms of strawberries. This is the opportunity cost of the choice made due to scarce resources such as land and labour.